

Linear Algebra Bootcamp 2

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Claim 9a. Show $\text{Hom}_F(F^m, F^n) \cong M_{n \times m}$.

Proof. Let $\alpha : \text{Hom}_F(F^m, F^n) \rightarrow M_{n \times m} \ni \phi \rightarrow \begin{bmatrix} \begin{array}{c} | \\ \phi(e^1) \\ | \end{array} & \cdots & \begin{array}{c} | \\ \phi(e^m) \\ | \end{array} \end{bmatrix}$

Define $\beta : M_{n \times m} \rightarrow \text{Hom}_F(F^m, F^n)$, where for all $A \in M_{n \times m}$ where $A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}$, $\beta(A) = [f \rightarrow \sum_{i=1}^n a_i \cdot f_i]$.

$\forall A \in M_{n \times m}$, $(\alpha \circ \beta)(A) = \alpha(\beta(A)) = \alpha([f \rightarrow \sum_{i=1}^n a_i \cdot f_i]) = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix} = A$. Thus $\alpha \circ \beta = \text{id}_{M_{n \times m}}$.

$\forall \phi \in \text{Hom}_F(F^m, F^n)$, $(\beta \circ \alpha)(\phi) = \beta \left(\begin{bmatrix} \begin{array}{c} | \\ \phi(e^1) \\ | \end{array} & \cdots & \begin{array}{c} | \\ \phi(e^m) \\ | \end{array} \end{bmatrix} \right) = [f \rightarrow \sum_{i=1}^n f_i \cdot \phi(e^i)] = [f \rightarrow \phi(\sum_{i=1}^n f_i e_i)] = [f \rightarrow \phi(f)] = \phi$. Thus $\beta \circ \alpha = \text{id}_{\text{Hom}_F(F^m, F^n)}$.

Thus α and β are bijections and $\text{Hom}_F(F^m, F^n) \cong M_{n \times m}$. □

Claim 9b. Explain why $\text{Hom}_F(F^m, F^m)$ is a an F -algebra (a “ring” with a natural copy of F inside of it).

Proof. By 9a, $\text{Hom}_F(F^m, F^m) \cong M_{m \times m}$. $F \subset M_{m \times m}$ through the injective map $f \rightarrow fI$. $M_{m \times m}$ forms a group under matrix addition. Then $GL_m(F) \subset M_{m \times m}$ forms a group under matrix multiplication. All elements of $M_{m \times m} \setminus GL_m(F)$ do not have a multiplicative inverse. Thus $M_{m \times m}$ is a ring with a natural copy of F inside it. □

Claim 10. Let $A : V \rightarrow W$ be a linear map. Show that we can find a basis for V and a basis for W so that the resulting matrix has one block I_k for some k , and everywhere else zero.

Proof. Let $m := \dim V$ and $n := \dim W$. By problem 7, $F^m \cong V$ and $F^n \cong W$. Let $\{v_1, \dots, v_m\}$ be some basis of V and $\{w_1, \dots, w_n\}$ be some basis of W . Let $f : V \rightarrow F^m$ be the iso such that $v_i \rightarrow e^i$. Let $g : F^n \rightarrow W$ be the iso such that $e^i \rightarrow w_i$.

Let $B := F^n \rightarrow F^m$ the resultant matrix such that $A = g \circ B \circ f$. Elementary column and row operations and permutations can be represented as invertible matrices. So let R^{-1} be the matrix corresponding to all the row operations of Gaussian Elimination and let C^{-1} be the matrix corresponding to all the column operations of gaussian Elimination. By construction there exists a matrix D which is the identity matrix flanked by 0s. Thus $D = R^{-1}BC^{-1} \implies B = RDC$. Thus $A = g \circ R \circ D \circ C \circ f$. R and D are invertible matrices and hence isomorphisms. Thus $g \circ R$ is selecting a basis of W and $C \circ f$ is selecting a basis of V . □

Claim 11. *Prove that the row space and column spaces of $n \times m$ matrices are the same.*

Proof. By the previous problem, $\forall A \in M_{n \times m}$ we can find a R and C such that $RAC = D$ where $D = \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix}$. Since R and C are invertible matrices, they don't change the number of independent rows or columns. Every row or column which has a 1 is linearly independent as they all have 1s in different positions of the tuple. Moreover by definition of I_k every row which has a 1, that column also has a 1 and vice versa. Thus the number of linearly independent rows and columns are equal which is k by definition of identity matrix. Thus there exists an isomorphism from F^k to the column space and row space of A . Thus they are the same. $\square$

Problem 12a. *When $V = F^n$, then A is an $n \times n$ matrix. how does it change when we change basis?*

Solution. When we change a basis, we apply an automorphism of the vector space. So if we change the basis of the domain, we first apply the automorphism then A . If we change the basis in the range space, we first apply A then apply the automorphism. However in practice applying a linear automorphism means multiplying by an invertible matrix. So if we change the basis in the domain and the range, it would be $\beta \circ A \circ \alpha$ where α is the automorphism which turns the new basis in the domain to the old basis in the domain and β is the automorphism which turns the old basis in the range to the new basis in the range. $\blacksquare$

Problem 12b. *Now we ask given $A : V \rightarrow V$ how simple can we make the matrix — or given an $n \times n$ matrix, what is the nicest matrix in its conjugacy class.*

Solution. The nicest and simplest matrix in its conjugacy class is the Jordan form of the matrix. $\blacksquare$

Problem 13. *Explain why it is not reasonable to hope that a matrix is conjugate to kI (identity multiplied by the scalar k).*

Claim 13. *A matrix is conjugate to a scalar matrix $\iff$ it itself is a scalar matrix.*

Proof. $\implies$: Let $A \in M_{n \times n} \ni \exists v \in GL_n(F) \ni vAv^{-1} = kI \implies A = v^{-1}kIv = kIv^{-1}v = kI$
 $\impliedby$: Let $A = kI$. Then $\forall v \in GL_n(F), vAv^{-1} = vkIv^{-1} = kI$. $\square$

Problem 14. *Show that the matrix for A , in some basis, is diagonal, $\iff$ the basis consists of Eigenvectors.*

Proof. ($\implies$): Suppose A is diagonalizable. Then there exists a $v : F^m \rightarrow V$ such that $D = v^{-1}Av$ where D is a diagonal matrix with scalars on the diagonal and 0s everywhere else. Thus $A = vDv^{-1}$. Let $w \in V$ be a vector such that $v^{-1}w = e_m^i$ is the i th standard basis of F^m . Thus $Aw = vDv^{-1}w = vke_m^i = kve_m^i = kw$. Since there are m standard basis vectors in F^m and v^{-1} and v are isomorphisms there are m possibilities for eigenvectors. We can find m linearly independent vectors that are eigenvectors thus they form a basis.

($\impliedby$): Suppose A has an eigenvector basis. Construct an isomorphism $v : V \rightarrow F^m$ which takes the i th eigenvector basis to the i th standard basis. Thus $v^{-1}e_m^i$ is the i th eigenvector basis. Matrix D is going to be a matrix such that $D_{i,i}$ is the eigenvalue for the i th eigenvector and $D_{i,j} = 0$ for all $i \neq j$. Thus $A = vDv^{-1}$ since v and v^{-1} are isomorphisms and D scales the eigenvector by its respective eigenvalue. Thus A is diagonalizable. $\square$

Problem 15a. Now for each f , define $V_f(A)$ to be those vectors v such that $Av = fv$. Show this is the same as the kernel of $A - fI$, in particular it is a sub space.

Proof. $\forall v \in V \ni Av = fv \implies Av - fv = 0 = (A - fI)v$ so by definition $v \in \ker(A - fI)$. Thus $V_f \subseteq \ker(A - fI) \forall v \in \ker(A - fI), (A - fI)v = 0 = Av - fIv = 0 = Av - fv = 0 \implies Av = fv$. Thus $\ker(A - fI) \subseteq V_f \implies V_f = \ker(A - fI)$. V_f is a subspace because the kernel of a linear map is subspace. $\square$

Problem 15b. Prove this is zero for all but finitely many f . Indeed show its non-zero iff $\det(A - fI) = 0$, and now produce a polynomial $P_A(X)$ so that f is an eigenvalue iff f is a root of this poly

Proof. $\implies : \forall f \in F \ni V_f(A) \neq 0$, by definition $\exists v \in V \setminus \{0\} \ni v \in V_f(A)$. Thus by definition, $Av = fv \implies (A - fI)v = 0 \implies \det(A - fI) = 0$.

$\impliedby : \forall f \in F \ni \det(A - fI) = 0 \implies \exists v \in V \setminus \{0\} \ni (A - fI)v = 0 \implies Av = fv \implies v \in V_f(A) \implies V_f(A) \neq 0$.

$P_A(x) := \det(A - xI)$ is the polynomial for which x is a eigenvalue iff it is the solution to the polynomial. P_A has a degree of at $\dim V$ because there are $\dim V$ x 's on the diagonal. By the fundamental theorem of algebra, a polynomial cannot have more solutions than the order of the polynomial. Thus the number of solutions or eigenvalues is at most $\dim V$. $\square$

Problem 16. Now consider $\oplus_{f \in F} V_f(A) \rightarrow V$. — the natural addition map. Explain why A is diagonalizable iff this map is an iso. show that, for any A , this map is always injective. Conclude A is diagonalizable iff $\sum_{f \in F} \dim V_f(A)$ is at least the dimension of V .

Claim 16a. The map is always injective.

Proof. Base case ($k = 2$): $\forall v_1 \in V_{f_1}$ and $\forall v_2 \in V_{f_2}$ where $f_1 \neq f_2$. Let:

$$v_1 + v_2 = 0 \tag{1}$$

Apply A to both sides:

$$Av_1 + Av_2 = 0$$

By definition of eigenspaces,

$$f_1 v_1 + f_2 v_2 = 0 \tag{2}$$

Multiply f_1 to (I):

$$f_1 v_1 + f_1 v_2 = 0 \tag{3}$$

Subtract (III) for (II):

$$(f_2 - f_1)v_2 = 0 \tag{4}$$

Since $f_2 \neq f_1$, thus $f_2 - f_1 \neq 0$. Thus $v_2 = 0$. Then $v_1 + v_2 = v_1 + 0 = v_1 = 0$. Thus $v_2 = v_1 = 0$.

Assume that $\sum_{i=1}^k v_i = 0$ with v_i in V_{f_i} with the f_i distinct implies $v_i = 0$ for all i .

Inductive Step: $\sum_{i=1}^{k+1} v_i$ with v_i in V_{f_i} with the f_i distinct.

$$\sum_{i=1}^{k+1} v_i = \sum_{i=1}^k v_i + v_{k+1}$$

By our assumption:

$$\sum_{i=1}^k v_i + v_{k+1} = 0 + v_{k+1} = v_{k+1} = 0$$

So, in the inductive step, we've shown that if $\sum_{i=1}^{k+1} v_i = 0 \implies v_i = 0$ for all i . Thus the kernel of $\oplus_{f \in F}$ is only the zero vector. Thus $\oplus_{f \in F} V_f$ must be an injective map. $\square$

Claim 16b. *Prove that A is diagonalizable iff the sum of the dim of V_f over all f is at least $\dim V$ and that if this holds then this sum is exactly $\dim V$.*

Proof. ($\implies$): Suppose A is diagonalizable. By problem 14, it has an eigenvector basis. Thus take its eigenvector basis E . Since E is a basis of V , subsets of E should be basis elements of its subspaces. Define $B_f = V_f \cap E$. The subset of E which is in V_f . Furthermore each element in E belongs to one eigenspace by definition. Thus for each $e \in E$, $\exists f \in F$ s.t. $e \in V_f$. Furthermore by the previous problem, the elements of distinct eigenspaces are not linearly dependent. Thus for each $e \in E$, $\exists$ a unique $f \in F$ s.t. $e \in V_f$. Since E is a basis of V , B_f must be a basis of V_f . Thus $\text{span}(B) = \dim V_f$. If $B_f = \emptyset$, then $V_f = \{0\}$ and thus $\dim V_f = 0$. Thus $\sum_{f \in F} \dim V_f = \sum_{f \in F} \dim B_f = \sum_{f \in F} |B_f|$. Since for each $e \in E$, $\exists$ a unique $f \in F$ s.t. $e \in V_f$, $\sum_{f \in F} |B_f| = |E| = \dim V$.

($\impliedby$): Suppose the sum of the dim of V_f over all f is at least $\dim V$. Denote B_f as the basis set of V_f for all $f \in F$. By the previous proof, elements in different eigenspaces are linearly independent. For most $f \in F$, $V_f = 0$ thus $B_f = \{0\}$. Moreover by problem 5, since there are at most $\dim V$ eigenvalues, there must also be at most $\dim V$ nonzero eigenspaces. Let $B = \bigcup_{f \in F} B_f \setminus \{0\}$. B will contain linearly independent vectors and will contain eigenvectors by definition. $\dim \text{span}(B) = \sum_{f \in F} \dim V_f \geq \dim V$ because we are taking the union of linearly independent basis sets from each V_f . By Linear Algebra Bootcamp 1 problem 7, we have shown that a dimension of a subspace must be less than or equal to its parent space and since $B \subseteq V$, then $\dim \text{span} B = \sum_{f \in F} \dim V_f \leq n$. Thus $n \leq \dim \text{span} B = \sum_{f \in F} \dim V_f \leq n$, or $\dim \text{span} B = \sum_{f \in F} \dim V_f = n$. Thus B contains n linearly independent vectors and spans V and contains only eigenvectors. Thus B is an eigenvector basis of V . By problem 4, A is diagonalizable. $\square$

Claim 16c. *A is diagonalizable iff this map is an iso*

Proof. $\implies$: A is diagonalizable. By claim 16b, the $\sum_{f \in F} \dim V_f = \dim V$. Thus the map is surjective. By claim 16a, the map is injective. Thus the map is an iso

$\impliedby$: If the map is an iso, the map is surjective, and the $\sum_{f \in F} \dim V_f = \dim V$. Thus by claim 16b, A is diagonalizable. $\square$